

RATCHANA POUROUCHOTTAMAN

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Master of AI graduate from Monash University with hands-on experience across machine learning, computer vision, NLP, and data science. I independently built a production RAG pipeline using LangChain, ChromaDB and LLMs, and led the AI development for a City of Melbourne sustainability project deploying a multi-model computer vision pipeline presented at a Monash industry event. I also designed and built a real-time smart parking platform integrating multiple open government datasets. My work has consistently been delivered to real deadlines and evaluated by academic and industry reviewers. Looking to apply my AI and data skills to problems that have genuine real-world impact.

PROJECTS

1. *Leafy Haven RAG Assistant (Personal Project)*

- Built a RAG-powered Q&A assistant extending the Leafy Haven rooftop greening application, enabling users to ask natural language questions grounded in a curated knowledge base of Melbourne greening guidelines.
- Implemented a full retrieval pipeline using LangChain RetrievalQA, ChromaDB vector store, and HuggingFace sentence-transformers (all-MiniLM-L6-v2) for semantic search across 31 document chunks.
- Deployed a live Gradio chat interface with source citation on every answer using Llama 3.1 8B via Groq API.

2. *Leafy Haven Rooftop Greening (City of Melbourne Sustainability Project)*

- Conceived the project idea and led all AI development within a team of 6, owning the pipeline from model selection to deployment.
- Built a multi-model computer vision pipeline integrating FastSAM for rooftop segmentation, GrabCut for mask refinement, and Stable Diffusion inpainting for green visualization, with a custom HSV-based vegetation detector to assess existing green coverage.
- Designed a green factor scoring model that translated raw model outputs into interpretable rooftop suitability scores across multiple environmental dimensions.
- Deployed as a fully functional web application and presented the AI architecture at the Monash Data Science & AI industry event to academic and industry reviewers.

3. *Smart Parking Availability System (Onboarding Project – City of Melbourne)*

- Designed and built the MySQL database schema to integrate 6 open government datasets including ABS census data, City of Melbourne parking bay sensors and LGA boundary data.
- Led the data insights epic, analysing car ownership and population growth trends across Melbourne to surface the root causes of CBD congestion.
- Contributed to a deployed web platform providing Melbourne commuters with real-time parking availability via an interactive map interface.

4. *Rift Valley Fever Clinical Trial Scheduling (MiniZinc)*

- Built a multi-objective constraint optimisation model in MiniZinc to schedule treatment allocations across patient groups, balancing total treatment cost against scheduling feasibility.
- Conducted a full sensitivity analysis across 9 datasets and 10 constraint groups using the HiGHS 1.9.0 solver, systematically removing each constraint to measure its impact on solution cost and runtime.
- Found that fixed vaccination frequency was the biggest cost driver, while cost-ratio constraints caused solver timeouts in complex scenarios showing ability to reason about computational trade-offs.

5. Mine Planning Optimization (MiniZinc)

- Built a MiniZinc optimisation model to find the most profitable underground mine tunnel path across a 2D grid, maximising ore yield while staying within budget and tunnel length constraints.
- Modelled complex spatial constraints including orthogonal adjacency, unstable cell reinforcement costs, and adjacent ore collection, handling both fixed and bounded tunnel length variants.
- Solved both problem variants (fixed and bounded length), with the bounded version requiring a more flexible model where unused tunnel positions are handled dynamically.

E D U C A T I O N

Master of Artificial Intelligence

Feb 2024 – Dec 2025

Monash University – Melbourne, VIC

Relevant Units: Machine Learning · Deep Learning · Discrete Optimisation · Planning & Automated Reasoning · Industry Experience Studio Project

Bachelor of Technology – Electronics & Communication Engineering Minor in Computer Science

Aug 2019 - May 2023

Puducherry Technological University

T E C H N I C A L S K I L L S

- **Core Language:** Python · SQL
- **Additional Programming:** C · C++ · JavaScript
- **ML & AI:** PyTorch · OpenCV · Ultralytics · Diffusers · Transformers · HuggingFace · LangChain · ChromaDB · RAG · LLM APIs
- **Optimisation:** MiniZinc · Constraint Programming · HiGHS Solver
- **Web & Backend:** FastAPI · HTML · CSS · JavaScript · Gradio
- **Databases:** MySQL
- **Security Tools:** Burp Suite · Wireshark · Kali Linux · Nmap · Hydra · Gobuster
- **Other:** Git · Linux · Google Workspace

W O R K E X P E R I E N C E

Chef & Crew – Guzman Y Gomez

Dec 2025 – Present

- Progressed from casual crew to full-time chef in a high-volume kitchen environment, trusted with opening and closing duties and recognised for reliability and operational ownership.
- Trained new staff members on food preparation standards, station procedures and food safety compliance.

HR Trainee – Tweezerman India Pvt. Ltd.

Jul 2023 – Dec 2023

- Supported HR operations including recruitment coordination, employee record management and administrative reporting.
- Handled sensitive employee information with confidentiality and accuracy.

I N T E R N S H I P

Cybersecurity and Ethical Hacking internship

Jul 2021 – Sep 2021

RINEX Technologies, in collaboration with E-Cell, IIT Kharagpur

- Completed hands-on penetration testing across two projects - a network hackathon and web application security assessment using industry-standard tools on Kali Linux.
- Performed network reconnaissance, port scanning, brute-force attacks and privilege escalation using Nmap, Hydra and Gobuster to identify and exploit vulnerabilities in a controlled environment.
- Conducted web application security testing identifying vulnerabilities including SQL injection, CSRF, clickjacking, sensitive data exposure and security misconfiguration using Burp Suite and Wireshark.